Project Title : Rice Leaf Disease Prediction System Report – Phase 4

Project video demonstration : CSCE 5214 Phase 4 Video.mp4

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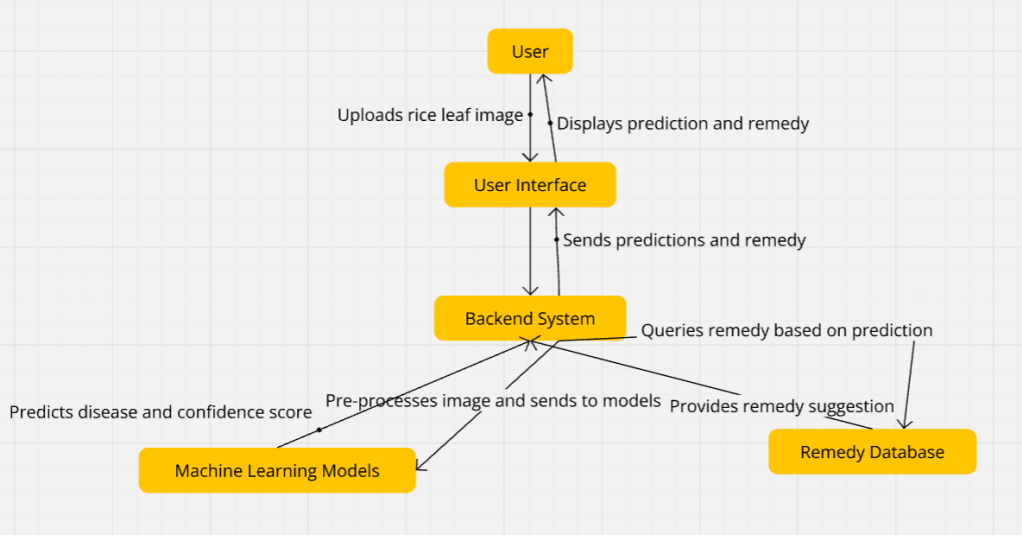
Phase 4: User Interface Development.

1. Introduction

During this stage, a user interface is designed to allow the rice leaf disease detection system to be user-friendly. The previous stages were dedicated to the model building and the system design, whereas this stage bridges that elements with a web-based interface. The idea is to enable users to post pictures and see results without engaging any code.

2. Communication Diagram Description

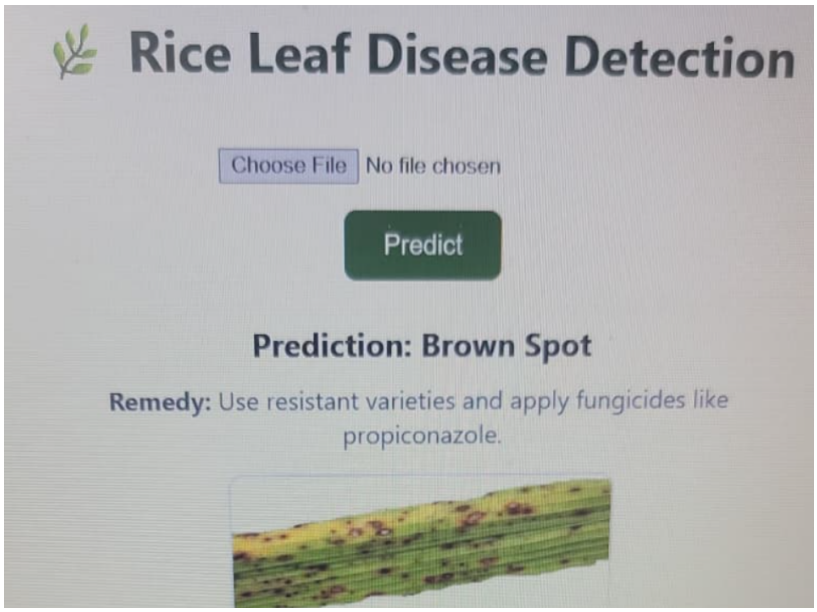
The communication diagram illustrates the interaction between different components of the system in the process of prediction.



The interface for uploading an image sends a request to the backend server when a user uploads an image. The backend analyses the image and sends it to the trained machine learning model. The model forecasts the kind of disease and sends the given outcome back to the backend. The backend then gets the correct remedy and forwards the prediction and remedy to the user interface. Lastly, the UI presents the results to the user in an easily understandable format.

Communication between user, interface, backend and model components flow smoothly through this flow.

3. User Interface Implementation



- The interface is developed using Flask along with HTML for structure. It offers an easy to view and easy to understand design to ensure that users have easy navigation with the system.

- The UI has a file upload facility whereby the users can upload an image of a rice leaf. After uploading the image, the user then clicks the predict button to begin the process. Then, the system will forward the image to the back-end to be processed.

- Once it predicts, the interface will show the name of the disease and a suggested remedy. Output is presented in a clear manner to ensure that users comprehend the output.

4. Backend and Model integrations.

- Flask is used to connect the user interface with the backend. The backend server takes control of the image uploaded by the user, where the image is preprocessed (resizing and normalizing). The processed image is then fed to the trained model to be predicted.

- The model sends the identified disease, and the backend associates it to the appropriate cure. The end result is subsequently relayed to the interface, and revealed to the user.

- This integration provides that all components are integrated to achieve a whole system.

5. Conclusion

Then the addition of user interface enhances usability of the system. It enables preferential interaction with the model and results can be obtained faster. This renders the system more feasible in real life applications, particularly, when the users lack technical expertise.

6. Future Improvements

Additional features that include ability to post multiple images, enhanced visualisation of the results and a dashboard that can be used to track the predictions can further enhance the system. It can also be improved in the interface design, which can be more interactive and look more attractive.